

HUJI Applicative Research Report — Healthcare — Week 35, 2026

6 high-potential paper(s) · Healthcare · Weekly · Week 35, 2026 · Generated August 24, 2026

This week's digest highlights groundbreaking advancements from Hebrew University poised for significant commercial impact across diverse sectors. We feature an innovative off-the-shelf cell therapy offering new hope for remyelination in neurological disorders, alongside a next-generation single-cell sequencing platform set to revolutionize precision diagnostics and drug discovery. Additionally, the digest includes novel small molecule drug candidates targeting inflammatory bowel disease and bacterial pathogens, a sustainable AgriTech solution for pest-resistant crops, and a breakthrough in synthetic chemistry for pharmaceutical building blocks.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Tamir Ben-Hur** — Neuroscience, Immunology, Clinical, Synthetic Biology
- **Oren Ram** — Genomics, Diagnostics, Drug Discovery, Clinical
- **Zvi Granot** — Drug Discovery, Immunology, Clinical
- **Osnat Malka** — AgriTech, Genomics, Synthetic Biology
- **Zackaria Nairoukh** — Drug Discovery
- **Emanuel Hanski** — Drug Discovery, Immunology, Proteomics, Clinical

HEALTHCARE — RESEARCH HIGHLIGHTS

1. Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders. 36/50

"Off-the-shelf stem cell therapy targets remyelination for multiple sclerosis and other neurological disorders."

Main researcher: Tamir Ben-Hur · tamir@hadassah.org.il.

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; The Department of Neurology, The Agnes-Ginges Center for Human Neurogenetics, Hadassah-Hebrew University Medical Center, Jerusalem, Israel. Electronic address: tamir@hadassah.org.il.

Published in Stem cell reports · 2026-06

ABOUT THIS RESEARCH

Researchers developed a clinical-grade astrocyte therapy derived from embryonic stem cells that helps the brain repair damaged myelin sheaths. The therapy works by clearing debris and creating a supportive environment that encourages the growth of new insulating layers for nerves.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

COMMERCIALISATION METRICS

Novelty

6/10

While the use of hESC-derived astrocytes is a sophisticated approach to remyelination, the 'bystander mechanism' and the focus on microglial debris clearance are established concepts in neuroregeneration, making this more of an incremental application of cell therapy than a groundbreaking scientific paradigm shift.

Commercial Potential

8/10

The therapy uses clinical-grade, scalable hESC-derived cells targeting a high-unmet-need market (demyelinating disorders) with demonstrated safety and a clear mechanism of action. The 'clinical-grade' status and scalability suggest it is already positioned for translational development rather than being purely theoretical.

Market Size

9/10

The therapy targets demyelinating disorders (such as Multiple Sclerosis), which represent a massive, high-value patient population with significant unmet medical needs. The mechanism's ability to address multiple facets of remyelination failure suggests a broad, multi-indication platform potential within the neurodegenerative market.

Tech Readiness

5/10

The technology has transitioned from basic lab discovery to 'clinical-grade' manufacturing and has demonstrated efficacy in animal models (lysolecithin-induced), but it remains in the preclinical validation stage rather than undergoing human clinical trials.

IP Strength

8/10

The development of a 'clinical-grade' hESC-derived astrocyte product (AstroRx) suggests a proprietary, scalable manufacturing process and a specific cell composition that is highly patentable. The multi-modal mechanism of action—addressing debris clearance and OPC differentiation—provides multiple layers of therapeutic defensibility and platform potential for various CNS disorders.

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2. sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method **35/50** for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells.

"Breakthrough single-cell sequencing technology unlocks hidden disease mechanisms and accelerates precision drug discovery."

Main researcher: Oren Ram

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, The Hebrew University, 9190401 Jerusalem, Israel.

Published in Nucleic acids research · 2026-04

ABOUT THIS RESEARCH

Researchers have developed a new single-cell sequencing technology called sc-rDSeq that captures a much wider range of RNA types, including non-polyadenylated RNAs, at a lower cost and higher density than current standards. By applying this to lung cancer, they demonstrated the ability to identify hidden cell subpopulations and drug-resistance mechanisms that traditional methods miss.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

COMMERCIALISATION METRICS

Novelty

8/10

The technology represents a significant platform innovation by enabling full-length, non-polyA RNA capture in a scalable droplet format, overcoming a major technical limitation of current industry standards like 10x Genomics. This capability to access the 'dark transcriptome' (ncRNA, histone RNA, etc.) provides a new dimension of biological insight that is highly relevant for precision oncology.

Commercial Potential

8/10

The technology offers a significant, platform-enabling improvement over industry standards (10x Genomics) by capturing non-polyA RNA and providing higher UMI density, which has clear commercial utility for drug discovery and precision oncology. Its scalability and cost-effectiveness position it as a highly licensable tool for large-scale clinical biomarker profiling.

Market Size

8/10

The technology targets the high-value single-cell sequencing market by offering a platform-enabling solution for comprehensive transcriptomics, which is essential for advanced drug discovery and personalized oncology. Its ability to capture non-polyA RNAs and alternative splicing at scale provides a significant competitive edge in the expanding precision medicine and biomarker discovery sectors.

Tech Readiness

4/10

The technology is currently at the proof-of-concept stage, demonstrated through successful validation in a laboratory setting using cancer cell lines. While it shows significant performance improvements over existing methods, it has not yet been transitioned into a standardized, commercially available kit or integrated into clinical diagnostic workflows.

IP Strength

7/10

The technology represents a potentially patentable platform innovation by optimizing droplet-based capture for non-polyA RNA, offering a clear technical advantage over industry standards like 10x Genomics. However, defensibility may be challenged by the crowded landscape of single-cell sequencing chemistries and the difficulty of protecting specific microfluidic or bead-based optimizations once widely adopted.

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3. Modulating immune toxicity in ulcerative colitis using a small molecule RAB27A inhibitor.

31/50

"First-in-class oral small molecule targets RAB27A, offering a novel therapeutic approach for ulcerative colitis."

Main researcher: Zvi Granot

Immunyx Pharma, JBP Minrav Building, Jerusalem, 91120, Israel.; Department of Developmental Biology and Cancer Research, Institute for Medical Research Israel Canada, Hebrew University Medical School, Jerusalem, 91120, Israel.

Published in Inflammatory bowel diseases · 2026 Aug 21

ABOUT THIS RESEARCH

Researchers have developed a small molecule inhibitor targeting the RAB27A protein to reduce harmful immune responses in ulcerative colitis. This approach aims to treat inflammatory bowel disease by controlling the secretion of inflammatory mediators from immune cells.

COMMERCIAL ANGLE

Development of a first-in-class oral therapeutic for autoimmune colitis and other inflammatory disorders driven by RAB27A-mediated exocytosis.

COMMERCIALISATION METRICS

Novelty

6/10

Targeting RAB27A for immune modulation is a scientifically sound approach, but small molecule inhibition of Rab GTPases is a known pursuit in oncology and immunology, making this an incremental application to UC rather than a groundbreaking discovery.

Commercial Potential

8/10

Targeting RAB27A with a small molecule offers a highly druggable modality for a significant unmet need in ulcerative colitis, providing a clear path toward a proprietary therapeutic product.

Market Size

8/10

Ulcerative Colitis is a chronic, high-prevalence condition with a multi-billion dollar global market and significant unmet need for targeted, low-toxicity therapies.

Tech Readiness

3/10

The title indicates early-stage drug discovery focused on a specific molecular target (RAB27A) for a disease state, which typically represents basic laboratory research (proof-of-concept) rather than a clinical-stage asset.

IP Strength

6/10

Small molecule inhibitors of specific protein targets like RAB27A are generally highly patentable via composition-of-matter claims. However, without a specific chemical scaffold or novel delivery mechanism described, the defensibility is limited to the target validation.

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4. Creating resistance to the whitefly *Bemisia tabaci* in cassava through RNAi-mediated targeting of multiple insect metabolic processes. 31/50

"RNAi-engineered cassava offers robust whitefly resistance, promising enhanced food security and sustainable agriculture."

Main researcher: Osnat Malka

Published · 2026-05

ABOUT THIS RESEARCH

Researchers developed a way to make cassava plants resistant to whiteflies, which are the primary carriers of devastating diseases in East Africa. They used RNA interference (RNAi) to target and disrupt essential metabolic processes within the insects.

COMMERCIAL ANGLE

Development of genetically engineered, pest-resistant crop seeds and RNAi-based biopesticides to reduce yield loss in global cassava production.

COMMERCIALISATION METRICS

Novelty

6/10

While targeting multiple metabolic pathways via RNAi is a robust strategy, host-induced gene silencing (HIGS) for pest control is a well-established technique; the novelty here lies primarily in the application to a specific crop-pest system rather than a fundamental scientific breakthrough.

Commercial Potential

7/10

The technology addresses a massive, billion-dollar agricultural pain point with a clear product path (transgenic resistant cultivars). However, the commercial potential is slightly tempered by the regulatory hurdles of GMOs in African markets and the necessity for multi-gene targeting to prevent insect resistance.

Market Size

8/10

The target market is vast and economically critical, with annual losses exceeding \$1.25 billion and direct impact on food security for millions in East Africa. The RNAi approach offers a scalable, platform-like solution to a 'super-abundant' pest affecting a primary global staple crop.

Tech Readiness

3/10

The research is in the early laboratory stage, focusing on the identification of candidate genes and the production of RNAi constructs, which precedes field testing and regulatory approval.

IP Strength

7/10

The use of multi-gene RNAi targeting specific metabolic pathways allows for strong composition-of-matter patents on the constructs and potential 'stacking' strategies to prevent resistance. However, defensibility is slightly tempered by the general nature of RNAi techniques and potential regulatory hurdles in agricultural biotech.

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5. Stereocontrolled entry to borylated piperidines & via stepwise dearomative carboboration of pyridines. 29/50

"Novel palladium catalysis enables efficient, stereocontrolled synthesis of key pharmaceutical building blocks."

Main researcher: Zackaria Nairoukh · z.nairoukh@mail.huji.ac.il.

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Published · 2026-08-17

ABOUT THIS RESEARCH

Researchers have developed a new chemical method to transform simple pyridine molecules into complex, functionalized piperidine structures with high precision. This two-step process uses palladium catalysis to ensure the new chemical groups are attached to specific locations and orientations.

COMMERCIAL ANGLE

This method provides a highly efficient way to synthesize complex nitrogen-containing building blocks that are essential for creating new pharmaceutical drugs.

COMMERCIALISATION METRICS

Novelty

7/10

The research presents a highly sophisticated stepwise approach to complex piperidine synthesis and proposes a novel Pd(IV) mechanistic pathway, though dearomative functionalization and carboboration are established themes in organic synthesis.

Commercial Potential

6/10

The methodology offers a highly valuable synthetic tool for creating complex piperidine scaffolds used in drug discovery, but it remains a mechanistic/synthetic improvement rather than a standalone product or platform. Its commercial value is primarily as a licensable process or a contribution to the toolkit of medicinal chemistry outsourcing services.

Market Size

6/10

The method targets substituted piperidines, which are ubiquitous motifs in the pharmaceutical industry, providing access to high-value building blocks. However, as a specific synthetic methodology rather than a platform for a novel therapeutic class, the immediate market is limited to the specialized medicinal chemistry reagent space.

Tech Readiness

3/10

This is fundamental methodology research focused on synthetic organic chemistry and mechanistic validation, representing early-stage laboratory discovery rather than a developed product or platform. While the ability to create substituted piperidines is chemically valuable, the work currently lacks the process optimization or application-specific validation required for higher TRLs.

IP Strength

7/10

The stepwise dearomative carboboration represents a novel synthetic methodology with high potential for patenting diverse chemical space, though its defensibility is tied to the broader scope of palladium-catalyzed dearomatization. The ability to create complex, stereocontrolled piperidines—a privileged scaffold in medicinal chemistry—provides significant industrial value and a strong foundation for protecting specific downstream medicinal applications.

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6. Group A Streptococcus host-pathogen dual crosstalk.

28/50

"Disrupting bacterial metabolism: a novel target identified for developing desperately needed anti-infective drugs."

Main researcher: Emanuel Hanski · emanuelh@ekmd.huji.ac.il.

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Published in Trends in microbiology · 2026-06

ABOUT THIS RESEARCH

Researchers have discovered that Group A Streptococcus bacteria hijack the human cell's stress response to force the production of the amino acid asparagine, which the bacteria then consume to fuel infection. This mechanism creates a metabolic bridge between host cell stress and bacterial virulence, mimicking similar pathways seen in cancer cells.

COMMERCIAL ANGLE

The discovery of the PERK-eIF2 α -ATF4-Asn axis provides a novel metabolic target for developing small-molecule inhibitors to starve bacterial pathogens or disrupt their ability to exploit host resources.

COMMERCIALISATION METRICS

Novelty

8/10

The research identifies a highly specific, previously unknown metabolic hijacking mechanism where an extracellular bacterium exploits a host stress pathway (PERK-eIF2 α -ATF4) to drive nutrient biosynthesis. This discovery of a direct 'crosstalk' circuit provides a novel, targetable metabolic vulnerability that parallels oncogenic pathways, offering significant potential for new antimicrobial strategies.

Commercial Potential

6/10

The identification of a specific metabolic axis (ATF4-Asn) provides a clear target for novel antimicrobial drug development, though the transition from mechanistic discovery to a viable therapeutic lead remains speculative.

Market Size

7/10

The research targets Group A Streptococcus, a major global pathogen with high clinical burden, but its most significant market value lies in the potential to translate the ATF4-Asn metabolic axis into oncology therapeutics. While the infectious disease application is substantial, the broader addressable market is expanded by the cross-sector relevance to cancer metabolism and therapy resistance.

Tech Readiness

2/10

The paper describes fundamental mechanistic biology and discovery of a signaling pathway in a laboratory setting. It lacks any mention of therapeutic intervention, drug development, or preclinical validation required to move beyond basic research.

IP Strength

5/10

The paper describes a fundamental biological mechanism (the ATF4-Asn metabolic axis) rather than a specific therapeutic molecule or engineered platform. While the discovery provides a targetable pathway, the core finding is a natural phenomenon which is difficult to patent without specific, non-obvious chemical or biological interventions.

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